MA 5755: Data Analysis & Visualization in R/Python/SQL

Week 5: Hierarchical Clustering, DBSCAN, and Cluster Evaluation

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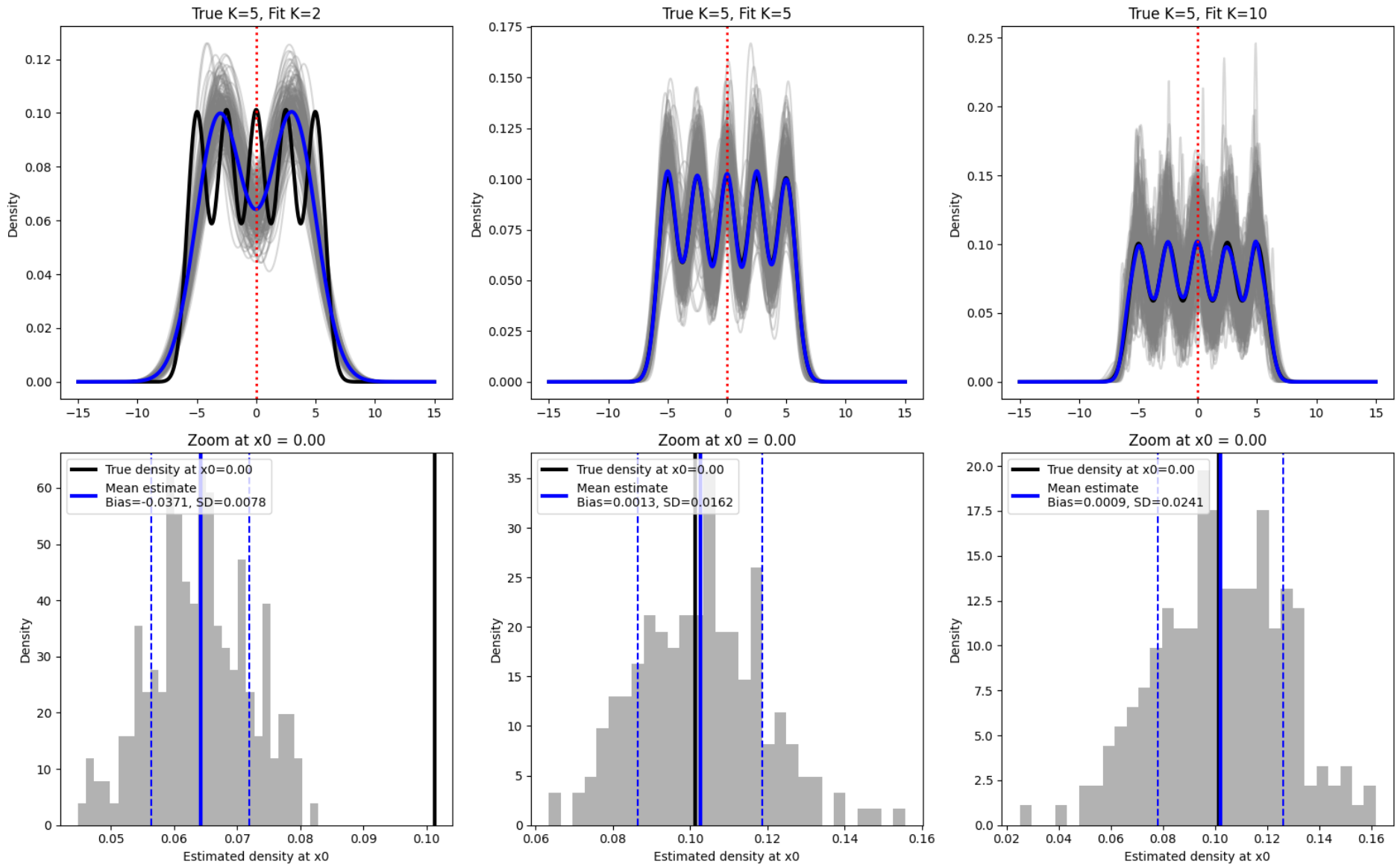
Outline

1. Review
2. Types of Clustering Methods
3. Hierarchical clustering
4. Density-based clustering
5. Evaluating Clusters

Bias-Variance Tradeoff for GMM



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Major Types of Clustering Methods

1. **Partition-based clustering**: e.g., k-means, k-medoids
   - Partition data into K disjoint clusters. The objective function minimization (e.g., within-cluster variance)
2. **Model-based clustering**: e.g., GMM
   - Assumes data generated from a probabilistic model where clusters correspond to mixture components
3. **Hierarchical clustering**: e.g., agglomerative, divisive
   - Produces a nested sequence of clusters represented as a tree (dendrogram)
4. **Density-based clustering**: e.g., DBSCAN
   - Clusters defined as high-density regions and based on connectivity and level sets of density

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Motivation for Hierarchical Clustering

Limitation of partition-based methods

- Methods such as k-means and k-medoids require:
  - Number of clusters K
  - Initial cluster assignment
- Results depend strongly on these choices

Hierarchical clustering: key idea

- Does not require pre-specifying number of clusters
- Requires only:
  - Pairwise dissimilarities between observations
  - A rule for measuring intergroup dissimilarity
- Produces nested sequence of clusterings

Hierarchy structure:

$$\text{Singleton clusters} \rightarrow \text{Intermediate clusters} \rightarrow \text{Single global cluster}$$

Agglomerative and Divisive Clustering

Two fundamental paradigms:

Agglomerative clustering (bottom-up)

- Start with N singleton clusters
- At each step merge two clusters with smallest intergroup dissimilarity
- Number of clusters decreases by one

$$N \to N - 1 \to \cdots \to 1$$

Divisive clustering (top-down)

- Start with one cluster containing all observations
- Recursively split clusters
- Split chosen to maximize between-group dissimilarity

$$1 \to 2 \to \cdots \to N$$

Both methods produce exactly $N - 1$ hierarchical levels.

Tree Representation of Hierarchical Clustering

Hierarchical clustering can be represented as a rooted binary tree.

Node interpretation

- Root node: entire dataset
- Terminal nodes: individual observations
- Internal nodes: clusters formed during merging or splitting

Agglomerative interpretation

- Parent node represents merged cluster
- Child nodes represent clusters that were merged

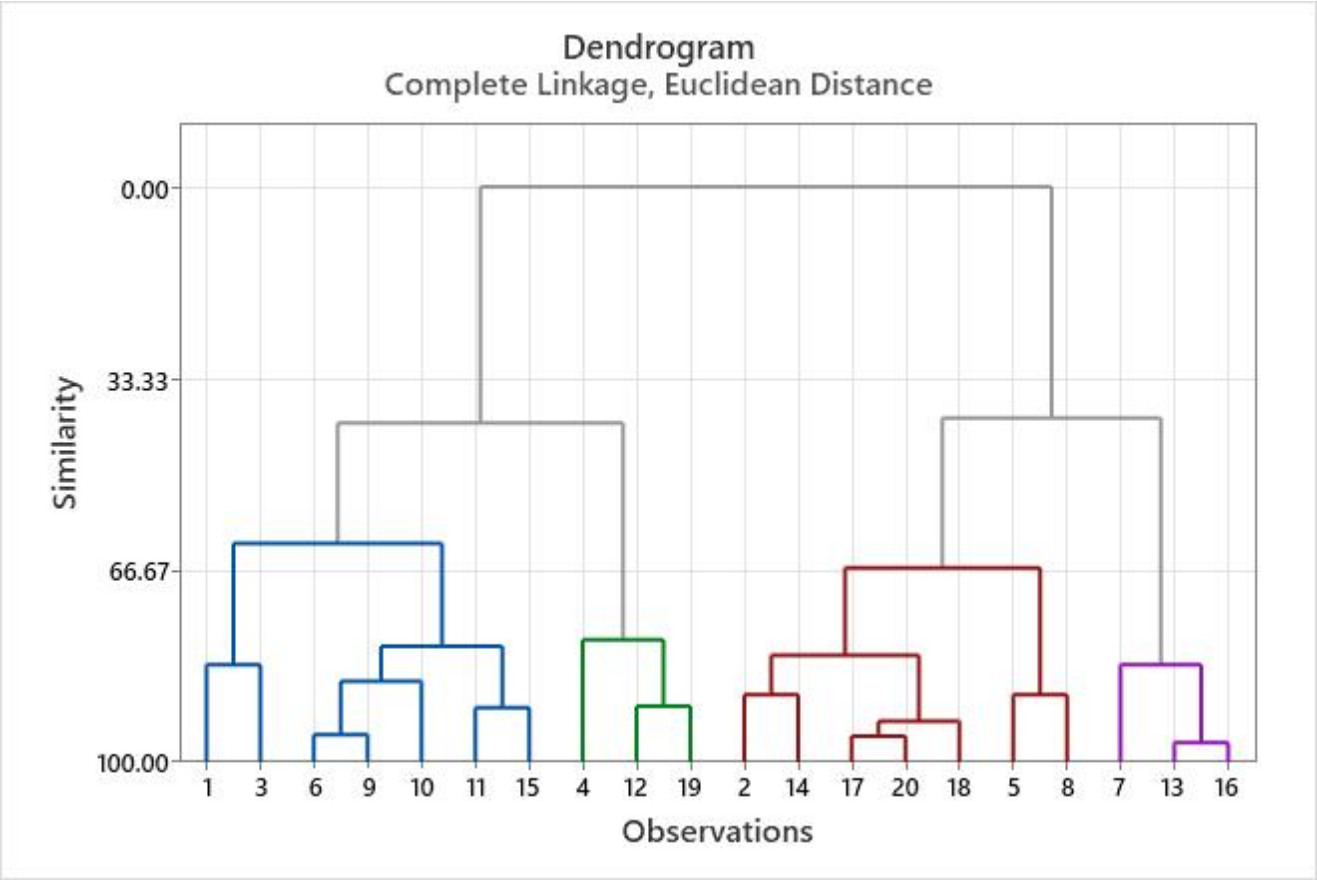
Divisive interpretation

- Parent node represents cluster before split
- Child nodes represent resulting subclusters

Tree represents entire sequence of clusterings.

Sample dendogram

A dendrogram provides a complete graphical description of the clustering hierarchy.



*Source: Minitab

Dendrogram Representation

Most agglomerative methods satisfy monotonicity property:

Intergroup dissimilarity increases with merge level

This allows graphical representation called a **dendrogram**.

Properties

- Leaves represent observations
- Internal nodes represent cluster merges
- Height represents intergroup dissimilarity

Cluster extraction

Cutting dendrogram at height h produces clusters.

- High cut $\rightarrow$ fewer clusters
- Low cut $\rightarrow$ more clusters

Dendrogram provides complete graphical description of clustering hierarchy.

Cophenetic Distance and Limitations

Cophenetic dissimilarity For observations (i, i'):

$$C_{ii'} = \text{dissimilarity at which they first join in same cluster}$$

Cophenetic correlation coefficient measures agreement between:

- Original dissimilarities $d_{ii'}$
- Cophenetic dissimilarities $C_{ii'}$

Ultrametric inequality

$$C_{ii'} \leq \max(C_{ik}, C_{i'k})$$

This imposes strong geometric constraints.

Important limitation

- Hierarchical methods impose hierarchical structure
- Even when true data structure is not hierarchical

Dendrogram should be interpreted cautiously.

Agglomerative Clustering: Basic Algorithm

Definition

Agglomerative clustering begins with each observation as a singleton cluster.

At each of the $N-1$ steps:

- Identify the two closest clusters
- Merge them into a single cluster
- Number of clusters decreases by one

This produces a hierarchy:

$$N \to N-1 \to \cdots \to 1$$

Key requirement: cluster dissimilarity

Let G and H be two clusters.

Cluster dissimilarity is computed from pairwise observation dissimilarities:

$$\{d_{ii'} : i \in G,\ i' \in H\}$$

Linkage Functions: Defining Cluster Dissimilarity

Three commonly used linkage methods:

Single linkage (nearest neighbor)

$$d_{SL}(G, H) = \min_{i \in G,\, i' \in H} d_{ii'}$$

Clusters are close if any pair of observations is close.

Complete linkage (farthest neighbor)

$$d_{CL}(G, H) = \max_{i \in G,\, i' \in H} d_{ii'}$$

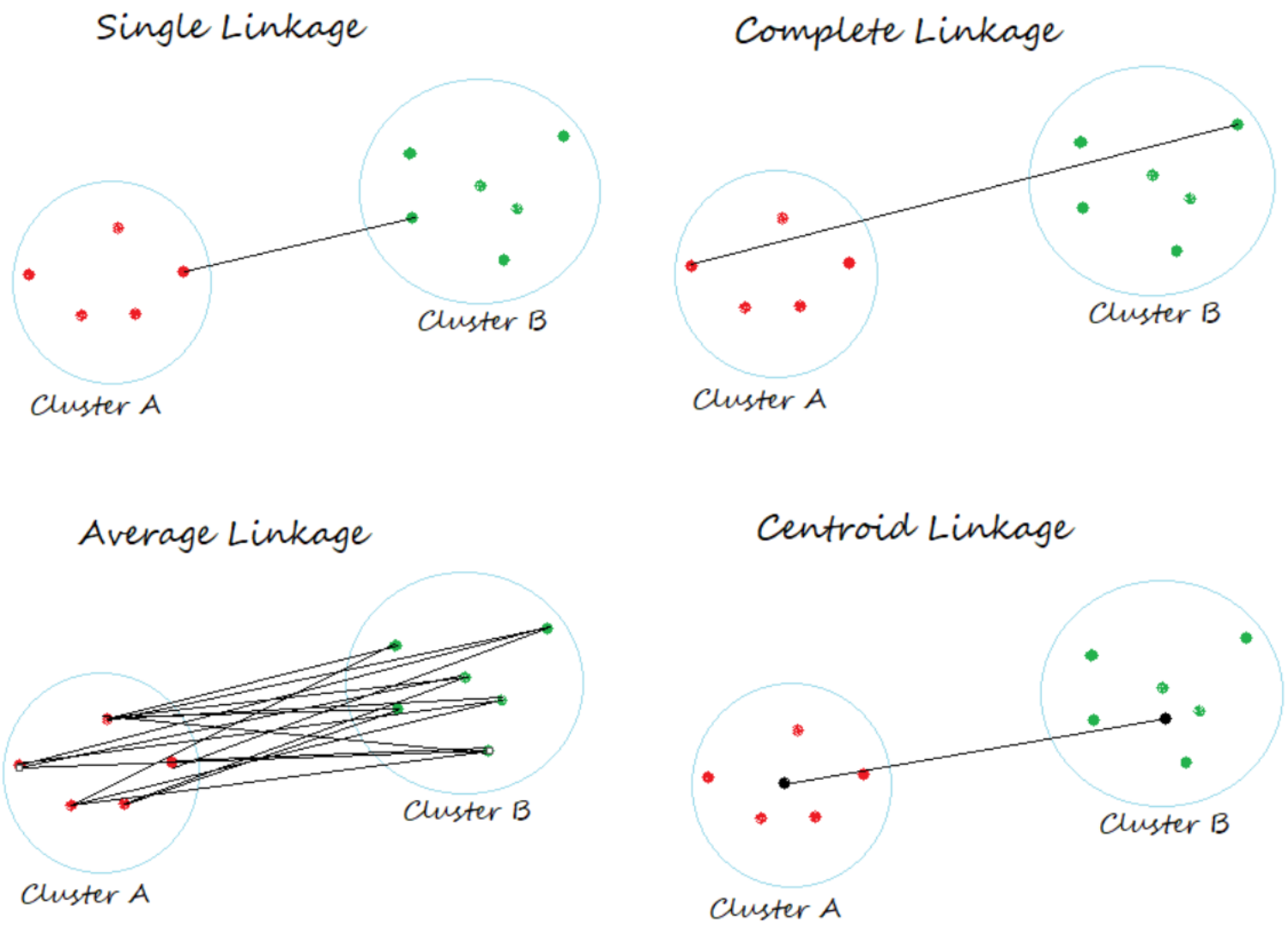
Clusters are close only if all observations are close.

Group average linkage

$$d_{GA}(G, H) = \frac{1}{N_G N_H} \sum_{i \in G} \sum_{i' \in H} d_{ii'}$$

Measures average dissimilarity between clusters.

Linkage Types



Cluster Compactness and Chaining Effect

Define diameter of cluster G:

$$D_G = \max_{i,i' \in G} d_{ii'}$$

Measures cluster compactness.

Single linkage properties

- Uses minimum distance between clusters
- Sensitive to chaining phenomenon
- Can produce clusters with large diameter
- May violate compactness property

Chaining occurs when clusters are connected by sequences of nearby observations.

Complete and Average Linkage Properties

Complete linkage properties

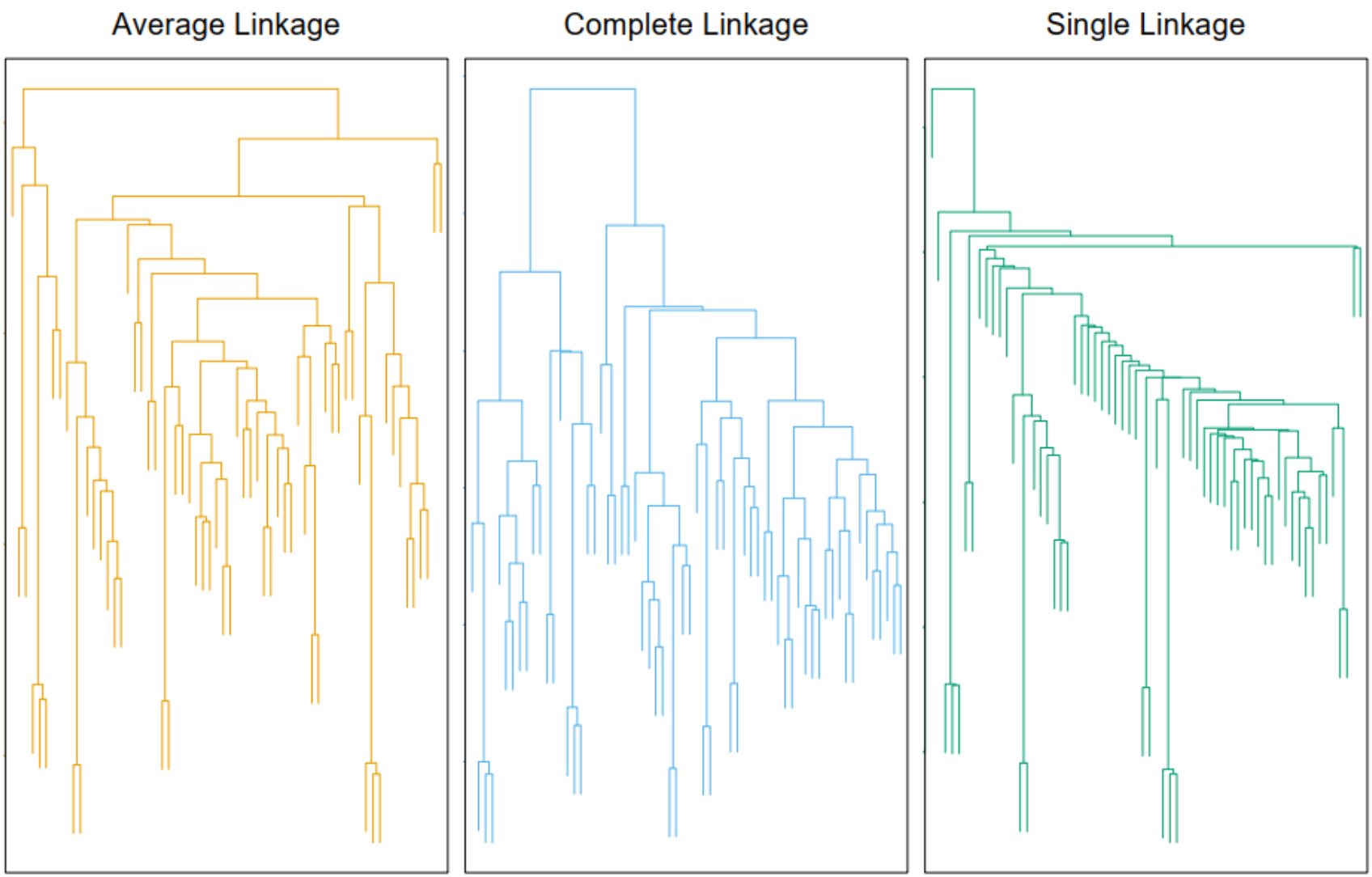
- Uses maximum distance between clusters
- Produces compact clusters
- Ensures small cluster diameter
- May violate closeness property

Observations in a cluster may be closer to another cluster than to some members of their own cluster.

Group average linkage

- Compromise between single and complete linkage
- Produces reasonably compact and separated clusters

Example



Example: Human Cancer Microarray Data

Hierarchical clustering was applied to gene expression microarray data.

Agglomerative clustering was performed using:

- Average linkage
- Complete linkage
- Single linkage

Observed results

- Average and complete linkage produced similar cluster structures
- Single linkage produced unbalanced clusters
- Single linkage exhibited chaining, producing long thin clusters

Average linkage provided the most interpretable clustering.

Advantages of Hierarchical Clustering in Microarray Data

Hierarchical clustering provides several advantages over K-means:

1. **Multi-resolution clustering**
   - Cutting dendrogram at different heights produces different clusterings
   - Clusters are nested within one another

$$\text{Fine clusters} \subset \text{Coarse clusters}$$

2. **Partial ordering of samples**
   - Dendrogram provides ordering of samples
   - Similar samples appear near each other

This reveals structure not visible in unordered data.

Divisive Clustering: Top-Down Approach

Divisive clustering begins with all observations in one cluster.

Recursive splitting produces hierarchy:

$$1 \to 2 \to 3 \to \cdots \to N$$

At each step:

- Select a cluster to split
- Divide into two daughter clusters

Possible splitting methods:

- K-means with $K = 2$
- K-medoids with $K = 2$

However, these methods may lack monotonicity required for dendrogram representation.

Macnaughton-Smith Divisive Algorithm

More principled divisive clustering method:

Start with cluster G containing all observations.

Step 1: Choose observation with largest average dissimilarity from others. This forms new cluster H.

Step 2: Iteratively transfer observations from G to H.
Transfer observation to H that maximizes (if positive):

$$\text{avg dissimilarity within } G - \text{avg dissimilarity to } H$$

Stop when no observation is closer to H than to G.

Repeat recursively to produce hierarchy.

Cluster selection for splitting based on:

- Largest diameter, or
- Largest average dissimilarity

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DBSCAN: Density-Based Spatial Clustering

DBSCAN = Density-Based Spatial Clustering of Applications with Noise

Key idea: Clusters are regions of high density separated by regions of low density.

Unlike k-means, DBSCAN:

- Does not require number of clusters K
- Can detect clusters of arbitrary shape
- Explicitly identifies noise points

Based on two parameters:

- ε (neighborhood radius)
- MinPts (minimum number of points required for dense region)

A point must have at least MinPts neighbors within distance ε to be considered part of a cluster.

Epsilon Neighborhood

Let (X, d) be a metric space.

Definition: ε-neighborhood
The ε-neighborhood of point x is:

$$N_\varepsilon(x) = \{y \in X : d(x, y) \leq \varepsilon\}$$

This defines the local region around point x.

Key idea:
Density at point x is measured by number of points in $N_\varepsilon(x)$.

Core, Border, and Noise Points

Points are classified into three types:

Core point

$$|N_\varepsilon(x)| \geq \text{MinPts}$$

Point lies in dense region.

Border point

- Not a core point
- Lies within ε-neighborhood of a core point

Noise point

- Neither core nor border
- Lies in low-density region

Density Reachability

Direct density reachability

Point y is directly density-reachable from x if:

- $y \in N_\varepsilon(x)$
- x is a core point

Density reachability

Point y is density-reachable from x if there exists sequence:

$$x_1, x_2, \ldots, x_n$$

such that each point is directly density-reachable from previous point.

This defines connectivity through dense regions.

Density Connectivity and Cluster Definition

Density connectivity

Points x and y are density-connected if:

$$\exists z \text{ such that both } x \text{ and } y \text{ are density-reachable from } z$$

Cluster definition

A cluster is a maximal set of density-connected points.

Noise points are those not belonging to any cluster.

DBSCAN Algorithm

Algorithm steps:

1. Select arbitrary unvisited point x
2. Compute $N_\varepsilon(x)$
3. If $|N_\varepsilon(x)| \geq \text{MinPts}$:
   - Start new cluster
   - Add all density-reachable points (essentially, add all neighbors of x to the cluster. If neighbor is a core point, add its neighbors too.)
4. Else mark x as noise
5. Repeat until all points processed

Produces density-connected clusters.

Advantages and Limitations of DBSCAN

- Does not require number of clusters
- Detects clusters of arbitrary shape
- Robust to noise and outliers
- Based on geometric and topological structure

Limitations

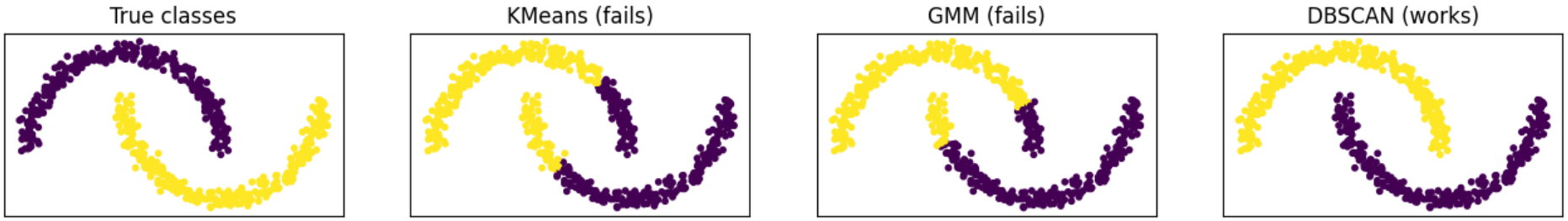
Performance depends on parameter selection:

- ε too small $\rightarrow$ many noise points
- ε too large $\rightarrow$ clusters merge
- MinPts affects density threshold

Less effective when densities vary significantly.

Computational complexity $O(n \log n)$.

Example and key things we will explore



Things we will explore in lab

- **Effect of density differences** Examine how DBSCAN behaves when clusters have different densities, and how fixed parameters (ε, MinPts) affect the result.
- **Density-bridging between clusters** Observe how clusters that are visually separate may still be merged if connected by chains of nearby points.
- **Effect of parameter choice** Explore how changing ε and MinPts alters the number and shape of clusters detected.

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Evaluating Clustering: Intrinsic Measures

Intrinsic measures evaluate clustering using only the data and cluster assignments. Used when true labels are **not available**.

Silhouette score

- Measures how similar a point is to its own cluster compared to other clusters; Range: $[-1, 1]$
- Higher values indicate better clustering

Davies–Bouldin index

- Ratio of within-cluster dispersion to between-cluster separation
- Lower values indicate better clustering

Calinski–Harabasz index (Variance Ratio Criterion)

- Ratio of between-cluster variance to within-cluster variance
- Higher values indicate better-defined clusters

Used for model comparison, parameter tuning, and validation.

Evaluating Clustering: Extrinsic Measures

Extrinsic measures evaluate clustering using known ground truth labels. Used when true class labels are available.

F-score (F-measure)

- Based on precision and recall
- Measures agreement between clustering and ground truth
- Higher values indicate better clustering

Rand index

- Measures agreement between true and predicted labels; Range: $[0, 1]$
- Higher values indicate better clustering

Variation of Information

- Measures difference between two clusterings using information theory
- Lower values indicate more similar clusterings

Fowlkes–Mallows Index: Definition

The **Fowlkes–Mallows index (FM)** is an extrinsic measure used to compare a clustering with ground truth labels.

It is based on pairwise comparisons between data points.

Define:

- TP: # pairs in same cluster in both true and predicted clustering
- FP: # pairs in same cluster in predicted clustering but not in true
- FN: # pairs in same cluster in true but not in predicted clustering

Definition

$$\text{FM} = \sqrt{\frac{TP}{TP + FP} \times \frac{TP}{TP + FN}}$$

This is the geometric mean of precision and recall.

Fowlkes–Mallows Index: Interpretation

Range

$$0 \leq \text{FM} \leq 1$$

- $\text{FM} = 1$: perfect agreement with true clustering
- $\text{FM} = 0$: no agreement
- Higher values indicate better clustering

Interpretation

- Measures how well clustering preserves pairwise relationships
- Penalizes both incorrect merges (FP) and incorrect splits (FN)
- Balanced evaluation using geometric mean

Use cases

- Comparing clustering algorithms
- Evaluating clustering accuracy when true labels are available
- Common in benchmarking clustering performance